

RESEARCH ARTICLE

Bi(III) Ion Electroreduction in the Presence of Ethanol and Sodium 1-Octanesulfonate at a Thin-Layer Amalgam Electrode (Renewable Liquid Silver Amalgam Film Electrode); Kinetic and Thermodynamic Aspects

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ABSTRACT

This study investigates the Bi(III) ions electroreduction in aqueous chlorate(VII) and aqueous-ethanolic chlorate(VII)-supporting electrolytes (SE) in the presence of the anionic surfactant sodium 1-octanesulfonate (1OSASS) over a range of temperatures. Electrochemical techniques, including direct current (DC) polarography, cyclic voltammetry (CV), and square-wave voltammetry (SWV), were used to evaluate both kinetic and thermodynamic parameters. Increasing ethanol concentration led to a decrease in the standard heterogeneous electron transfer rate constant (k_s), despite an increase in the diffusion coefficient of the oxidised species (D_{ox}). Raising temperature and surfactant concentration caused a shift in the formal potential (E_f^0) and an increase in redox peak separation (ΔE), indicating reduced process reversibility. Activation parameters, including activation energy (E_a), enthalpy (ΔH^0), entropy (ΔS^0), and Gibbs free energy (ΔG^0), were determined using the Arrhenius and Eyring equations. The results confirm that 1OSASS exhibits a ‘cap-pair’ effect, likely due to competitive adsorption and complexation, significantly affecting redox kinetics. Overall, the study highlights the complex influence of the solvent–surfactant system on the structure of the electrical double layer (EDL) and on the kinetics and mechanism of the Bi(III) ions electroreduction process.

1 | Introduction

The electroreduction of bismuth ions from a trivalent state to metallic bismuth (Bi(0)) is an important area of research because of the wide range of possible uses for bismuth and its compounds. These include protective coatings, electrochemical materials, optoelectronic devices, energy conversion and storage systems, and applications in medicine and diagnostic imaging [1, 2], semiconductor electronics [3, 4], and composite materials [5]. These diverse applications are mainly due to bismuth’s unique physicochemical properties. Its low toxicity, relatively high electrical conductivity, and resistance to oxidation in acidic environments make it a highly attractive material for anti-corrosive technologies. The use of this material extends beyond chemically passive coatings. Research has shown that adding Bi(III) ions to Ti/Sb/SnO₂/PbO₂ electrodes enhances surface hydrophobicity, boosts

current efficiency, and improves performance in the oxygen evolution reaction (ORR), as well as prolongs the lifespan of electrochemical coatings [6]. These findings highlight the role of bismuth as an effective modifier in industrial applications, including zinc electroplating processes [6]. Particularly noteworthy are ‘bismuthene’ structures- mono- and bilayer atomic sheets of Bi obtained via electrochemical synthesis. These structures show efficiency in reducing carbon dioxide CO₂RR, positioning them as potential materials for energy conversion systems [7]. In addition, metallic bismuth layers have gained notable interest as functional components in the development of modern electrochemical sensors. Electrodes containing Bi are widely used for detecting toxic metal ions, such as Pb²⁺, Cd²⁺, and Zn²⁺. Furthermore, when this element is integrated into porous carbon-based frameworks, its analytical performance is often

significantly improved. Notable examples include Bi–CX and Bi–CA composites. These materials have enabled the detection of trace levels of Pb^{2+} and Cd^{2+} , with detection limits as low as 0.02–0.16 ppb and highly responsive stripping signals [8, 9]. Additionally, multifunctional Bi–Fe carbon xerogels have shown the capacity to detect both heavy metals and biologically relevant substances (such as H_2O_2) at thresholds well below the limits set for drinking water; however, it should be noted that in these systems, the Fe content and structural features notably influence the electrochemical response [10]. These findings demonstrate that the analytical performance of electrodes based on bismuth, or containing bismuth as a key component, largely depends on the characteristics of the deposited metallic phase, including its stability, porosity, and dispersion. A parallel line of research focuses on the effects of chemical modifiers, such as surfactants, amino acids, and heterocyclic compounds, on the mechanism and kinetics of metal-ion electroreduction, including Bi(III), typically studied in chlorate(VII)-based solutions. One notable example is 6-thioguanine. It is a heterocyclic purine derivative that fulfils the so-called ‘cap-pair’ rule [11]. This rule describes the ability of an organic molecule to exert catalytic or inhibitory effects by forming transient active complexes with depolariser ions [12]. It has been shown that 6-thioguanine accelerates Bi(III) electroreduction. This idea aligns with Frumkin’s viewpoint, which emphasises how the progression of electrode reactions is influenced not only by the distribution of the electric field but also by specific adsorption and chemical interactions occurring within the electrical double layer (EDL) [13, 14]. Although the reaction $\text{Bi}^{3+} + 3\text{e}^- \rightarrow \text{Bi}^0$ represents the fundamental stoichiometric transformation, numerous studies have demonstrated the mechanistic complexity and multistep nature of this process. In recently conduct, experiments, focused on the step reduction of Bi(III) were carried out using both a hanging mercury drop electrode (HMDE) and a thin, renewable liquid silver amalgam film electrode (known as R-AgLAFe), as detailed in previous literature sources [15, 16]. System factors significantly impact the behaviour of electrode reactions. These factors include the electrode potential [17], the composition and polarity of the surrounding medium [18], and the influence of ligands that alter the interface between the electrode and the solution [19]. Despite the literature on Bi(III) electroreduction phenomena, there is still a lack of studies that combine the influence of solvent polarity, surfactant–depolariser interactions at the electrode/solution interface, and temperature-dependent kinetic-thermodynamic analysis. The authors found no reports on the study of Bi(III) reduction in mixed water-ethanol environments using chlorate(VII) and a renewable thin-film amalgam electrode (R-AgLAFe) or similar electrodes. Moreover, the polarity-dependent behaviour of the anionic surfactants in the Bi(III)/Bi(0) system has not yet been characterised, particularly with respect to transitions between inhibitory and catalytic regimes that extend the interpretation of the ‘cap-pair’. These aspects determine the innovative nature of this research. In this context, the present study aims to analyse the Bi(III) ions electroreduction in aqueous (chlorate(VII)) and aqueous-ethanolic (chlorate(VII)-ethanol) supporting electrolyte (SE) solutions in the presence of the anionic surfactant sodium 1-octanesulfonate (1OSASS), as a function of temperature, using selected electrochemical techniques. The results may contribute to a better understanding of how medium properties influence electrode processes and clarify how solution composition controls electron-transfer processes

relevant to controlled bismuth deposition and applications, among others, in electrochemical sensors.

2 | Experimental

2.1 | Chemicals and Reagents

SE solutions were prepared with 0.5 mol/dm^3 sodium chlorate(VII) (NaClO_4 , Fluka, $\geq 98.0\%$) combined with 0.5 mol/dm^3 chloric acid(VII) (HClO_4 , Chempur, 70%, analytical grade) to form system A. All compounds are named according to IUPAC recommendations; for example, chlorate(VII) (commonly known as perchlorate) and chloric acid(VII) (perchloric acid). To form mixed aqueous-organic SEs, water was replaced with ethanol (analytical grade, POCh), resulting in mixed systems containing 25 vol% (system B) and 50 vol% (system C) ethanol, respectively. 1OSASS (Fisher Scientific, $\geq 98\%$), an anionic surfactant, was added to the prepared solutions at concentrations ranging from $5 \cdot 10^5$ to $1 \cdot 10^3 \text{ mol/dm}^3$. The concentration of Bi(III) ions, introduced as bismuth(III) nitrate pentahydrate ($\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$, Sigma–Aldrich, $\geq 99\%$), was set to $1 \cdot 10^{-3} \text{ mol/dm}^3$ in all solutions.

2.2 | Instrumentation and Electrochemical Setup

The experiments were conducted using a standard three-electrode setup, in which the working electrode was a thin-film R-AgLAFe. This electrode offers a key practical advantage over classical HMDE systems, namely a significantly reduced mercury consumption. Depending on the measurement conditions and the composition of the medium, its operation requires only $\sim 20 \mu\text{L}$ of mercury for approximately 1200 film renewals (which should be regarded as an operational estimate rather than a fixed value), making R-AgLAFe a considerably more sustainable alternative. Additional benefits of this electrode include a wide potential window in chlorate(VII) media (from approximately +0.1 to -2.0 V), a highly reproducible amalgam surface regenerated smoothly and automatically, and precise control over the transfer time of the working electrode from the sensor body (amalgamation chamber) to the test solution [15, 16].

The surface area of the R-AgLAFe electrode ($0.17 \pm 0.01 \text{ cm}^2$) was calculated by defining its geometry as a cylinder and summing both the lateral surface area of the amalgam-coated segment and the area of the circular base that comes into contact with the solution.

The total area was therefore determined according to Equation (1):

$$A = \pi dl + \frac{\pi d^2}{4} \quad (1)$$

where d (the manufacturer-specified) denotes the wire diameter and l the length of the amalgam-coated segment (measured before the measurements were taken), the sensor preparation procedure included mechanical polishing with P1200 abrasive paper, rinsing with distilled water, and refreshing the mercury solution in the amalgamation chamber, as outlined by Nosal-Wiercińska et al. [20]. An Ag/AgCl ($3 \text{ mol/dm}^3 \text{ KCl}$) electrode was used as the reference electrode, while a platinum wire coil served as the counter electrode. The Ag^+/Ag equilibrium potential of the Ag/AgCl ($3 \text{ mol dm}^{-3} \text{ KCl}$) lies far positive of the

potential window used in this work and therefore cannot interfere with the recorded Bi(III)/Bi(0) signals. Measurements were carried out at isothermal conditions within a temperature range of 288–303 K (resolution 0.01 K, stability ± 0.2 K) using a μ Autolab type III/GpES potentiostat (EcoChemie B.V., Utrecht, the Netherlands).

The following electrochemical techniques were employed:

- Direct current polarography (DCP)
- Cyclic voltammetry (CV)
- Square-wave voltammetry (SWV).

3 | Results and Discussion

3.1 | Effect of Ethanol on the Electrode Process as a Function of Temperature

Investigations by Lipkowski and Galus, among others, focused on the electroreduction of metal ions in mixed aqueous-organic systems, showing that the addition of aliphatic alcohols to aqueous mixtures has a pronounced effect on the kinetics of electrode reactions. This effect is mainly due to alterations of the dielectric, viscosity, and ion transport within the EDL [21, 22]. Pursuing this direction, the Bi(III) ions electroreduction was analysed in SEs containing 0%(v/v), 25%(v/v), and 50%(v/v) ethanol. Heating the solution systematically increased the peak currents and the slope of DC polarographic waves relative to temperature for $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(III) ion electroreduction (Scheme 1). This behaviour is consistent with the Ilkovič equation, which predicts a proportional increase in the diffusion current with increasing diffusion coefficient [23, 24] (Table 1).

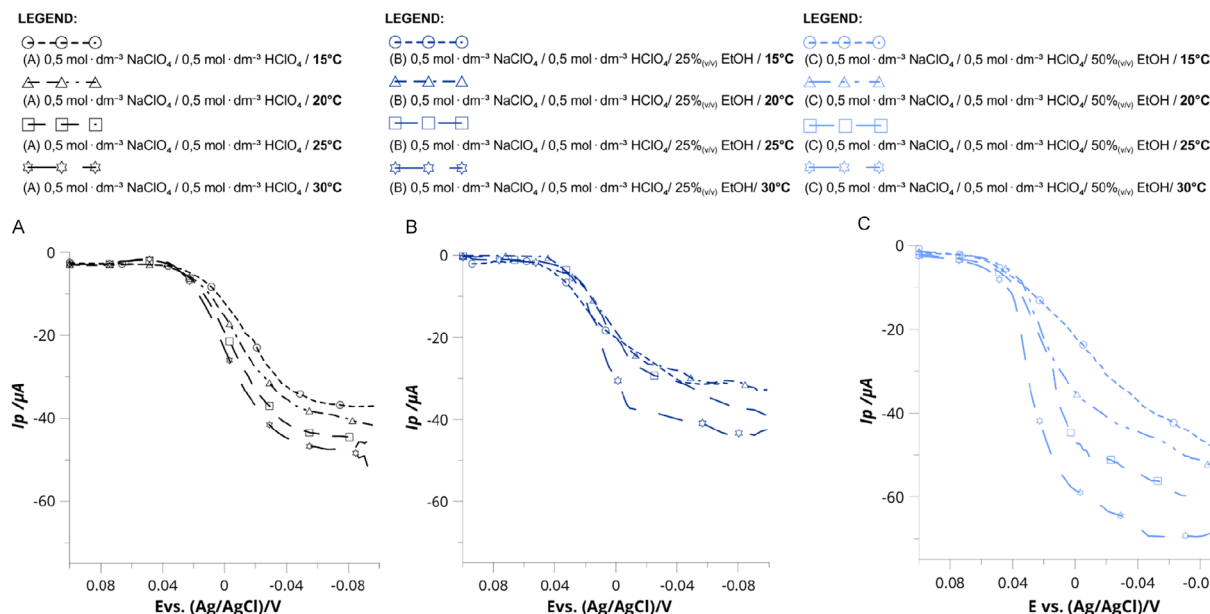
According to the Ilkovič equation:

$$i^d = K \cdot n \cdot D^{\frac{1}{2}} \cdot m^{\frac{2}{3}} \cdot t^{\frac{1}{6}} \cdot c \quad (2)$$

where i^d is the diffusion current, n is the number of electrons exchanged, D is the diffusion coefficient of the depolariser, m is the mercury flow rate, t is the drop time, and c is the bulk concentration of Bi(III) ions.

At the same time, higher ethanol content in the SE resulted in a systematic decrease in the standard rate constant (k_s) and flattening of the wave profiles (Table 1). These effects can be attributed to changes in solution viscosity due to ethanol addition and to EDL reorganisation, both of which reduce electron transfer efficiency. Additionally, increased ethanol content may weaken Bi(III) ion solvation, thereby reducing the effectiveness of intermolecular interactions that facilitate charge transfer.

To confirm the trends observed in the DC measurements, SWV was employed. The resulting curves (Scheme 2) are consistent with the conclusions drawn from the DC technique (Scheme 1). The reduction peak observed near 0 V versus Ag/AgCl (Scheme 2) corresponds to the $\text{Bi}^{3+} \rightarrow \text{Bi}^0$ process occurring at the electrode R-AgLAFE/solution interface. No direct experimental confirmation of Bi(0) formation was obtained in this study, which represents a limitation of the present work; therefore, the proposed interpretation is based on electrochemical evidence supported by previous studies performed under comparable conditions [16, 19, 20]. The appearance of a distinct anodic peak further supports this interpretation, indicating that the reduction involves the deposition and subsequent oxidation (stripping) of metallic Bi at the electrode/solution interface (Supporting Information, Figure S1, Table S1). Notable shifts of the peaks toward more positive potentials, along with changes in peak symmetry, further confirm the influence of ethanol on the kinetics and mechanism of the redox process. The values of formal potentials (E_f^0) and reversible half-wave potentials ($E_r^{1/2}$) presented in Table 2 exhibit dependence on both temperature and solution composition. In all examined systems, these potentials shift toward more positive values with increasing temperature, which may indicate a facilitation of Bi(III) ions electroreduction. Simultaneously, the presence of



SCHEME 1 | DC polarographic waves for the $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(III) ions electroreduction in (A) aqueous and (B,C) mixed aqueous-organic SEs containing ethanol at the concentrations indicated in the legend. Experimental conditions: interval time, 6 s (equivalent scan rate 0.000325 V s $^{-1}$); initial potential, -1.0 V; end potential, -1.0 V; step potential, 0.00195 V (SI, Table S5).

TABLE 1 | Diffusion coefficients of the oxidised form of the depolariser (D_{ox}) and standard rate constants (k_s) for the $1 \cdot 10^{-3} \text{ mol dm}^{-3}$ Bi(III) ions electroreduction in SEs.

$T, \text{ K}$	$D_{\text{ox}}, \text{ cm}^2 \text{ s}^{-1}$	$k_s, \text{ cm s}^{-1}$	$\ln_{\text{kf}}$
0.5 mol dm^{-3} NaClO ₄ + 0.5 mol dm^{-3} HClO ₄ + $1 \cdot 10^{-3}$ mol dm^{-3} Bi(NO ₃) ₃ ·5H ₂ O (SE A)			
288	$9.28 \cdot 10^{-6}$	$1.10 \cdot 10^{-4}$	−9.11
293	$8.25 \cdot 10^{-6}$	$1.13 \cdot 10^{-4}$	−9.09
298	$1.45 \cdot 10^{-5}$	$1.26 \cdot 10^{-4}$	−8.98
303	$1.58 \cdot 10^{-5}$	$1.36 \cdot 10^{-4}$	−8.90
0.5 mol dm^{-3} NaClO ₄ + 0.5 mol dm^{-3} HClO ₄ + $1 \cdot 10^{-3}$ mol dm^{-3} Bi(NO ₃) ₃ ·5H ₂ O/25% EtOH (SE B)			
288	$7.81 \cdot 10^{-6}$	$9.43 \cdot 10^{-5}$	−9.27
293	$9.36 \cdot 10^{-6}$	$9.81 \cdot 10^{-5}$	−9.23
298	$1.09 \cdot 10^{-5}$	$1.06 \cdot 10^{-4}$	−9.15
303	$1.35 \cdot 10^{-5}$	$1.27 \cdot 10^{-4}$	−8.97
0.5 mol dm^{-3} NaClO ₄ + 0.5 mol dm^{-3} HClO ₄ + $1 \cdot 10^{-3}$ mol dm^{-3} Bi(NO ₃) ₃ ·5H ₂ O/50% EtOH (SE C)			
288	$1.67 \cdot 10^{-5}$	$8.46 \cdot 10^{-5}$	−9.38
293	$1.83 \cdot 10^{-5}$	$8.73 \cdot 10^{-5}$	−9.35
298	$2.06 \cdot 10^{-5}$	$9.34 \cdot 10^{-5}$	−9.28
303	$2.73 \cdot 10^{-5}$	$1.07 \cdot 10^{-4}$	−9.14

ethanol leads to a further shift in potentials, most notably in the solution containing 50% (v/v) EtOH, where $E_r^{1/2}$ reaches its highest value. This suggests that ethanol affects not only the kinetic parameters but also the stability of intermediate complexes formed during the redox process.

The formal potentials (E_f^0) were calculated using the modified Randles method [24]:

$$E_f^0 = \frac{1}{2} \cdot \left[\frac{E_{\text{pa}}}{4} + \frac{E_{\text{pk}}}{4} + \left(\left(\frac{E_{\text{pa}}}{4} + \frac{E_{\text{pk}}}{4} \right) - \left(\frac{E_{3\text{pa}}}{4} + \frac{E_{3\text{pk}}}{4} \right) \right) / (g-1) \right] \quad (3)$$

where:

$$g = \left(\frac{E_{3\text{pa}}}{4} - \frac{E_{3\text{pk}}}{4} \right) / \left(\frac{E_{\text{pa}}}{4} - \frac{E_{\text{pk}}}{4} \right) \quad (4)$$

The reversible half-wave potentials were calculated using:

$$E_{1/2}^r = (E_{\text{pk}} + E_{\text{pa}}) / 2 \quad (5)$$

The charge-transfer coefficients (α) were determined using the classical Galus relation [24]:

$$\alpha \cdot n_a = 1.857 \cdot RT / \left[F \cdot \left(\frac{E_{\text{pk}}}{2} - E_{\text{pk}} \right) \right] \quad (6)$$

where $\frac{E_{\text{pk}}}{2}$ is the potential at half-height of the cathodic peak.

The cathodic transfer coefficients (α) exhibit irregular variations that depend on both temperature and the composition of the SEs. The values range from 0.15 to 0.37, indicating a change in the reaction mechanism in aqueous-ethanolic solutions and suggesting the involvement of different transition states or configurations of active complexes in the electrode process (Table 2).

Based on the $\ln k_s$ versus $1/T$ plots for each SE system (Scheme 3), the activation energies (E_a) for the electroreduction of Bi(III) ions were determined using the Arrhenius relation:

$$k_s = A \cdot \exp(-E_a/RT) \quad (7)$$

In linearised form:

$$\ln k_s = \ln A - E_a/(RT) \quad (8)$$

so that E_a is obtained from the slope of the $\ln k_s$ versus $1/T$ dependence [21, 22, 25].

The activation parameters were further evaluated using the Eyring–Polanyi equation [25, 26]:

$$k_s = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right) \quad (9)$$

The linearised form:

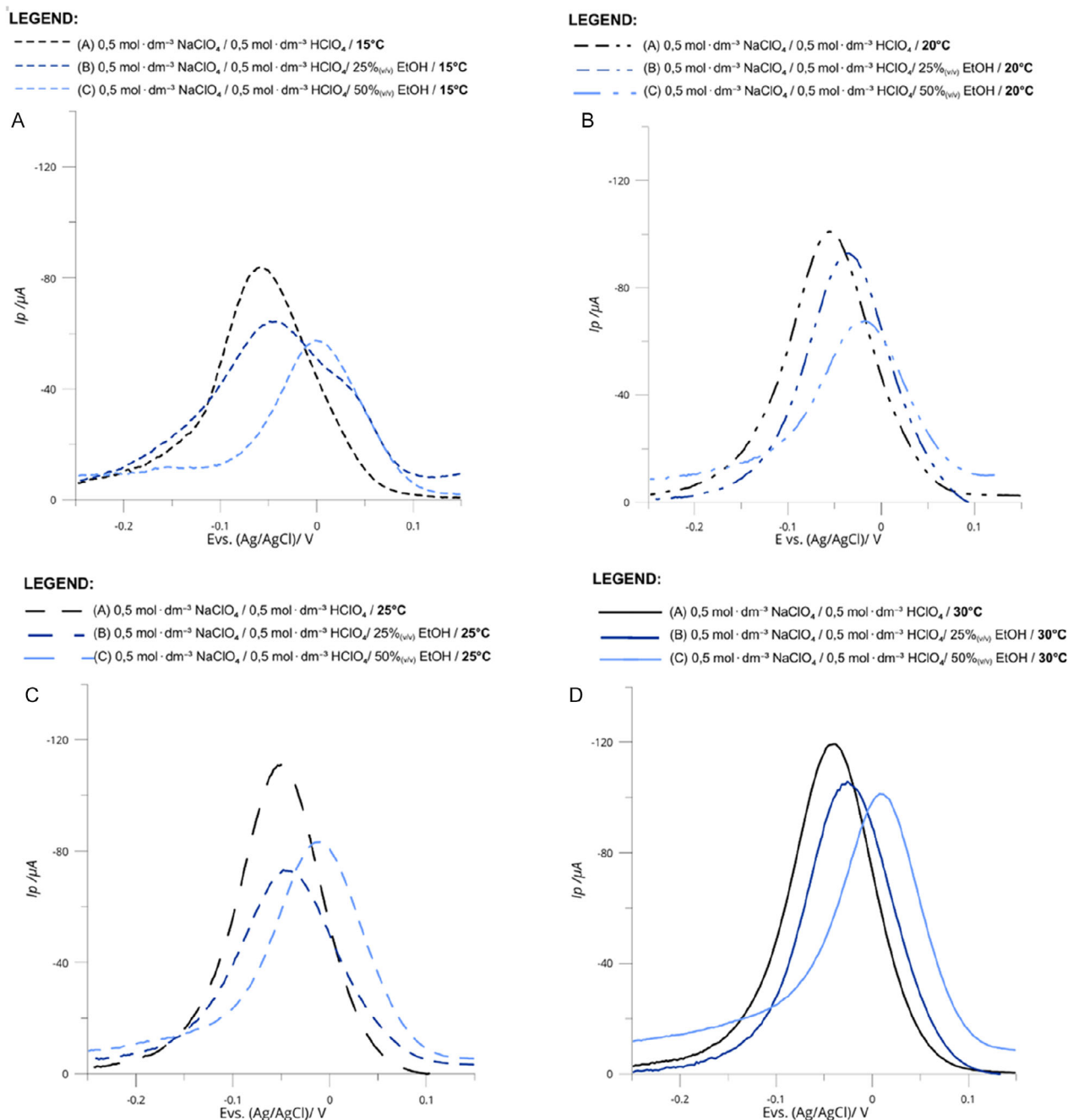
$$\ln\left(\frac{k_s}{T}\right) = -\frac{\Delta H^\ddagger}{RT} + \frac{\Delta S^\ddagger}{R} + \ln\left(\frac{k_B}{h}\right) \quad (10)$$

The Gibbs free energy of activation was calculated as:

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger \quad (11)$$

[25–27]. The results are summarised in Table 3.

The activation enthalpy values (ΔH) were, on average, approximately 2.6 kJ mol^{-1} lower than the activation energies (E_a) obtained from Arrhenius plots, consistent with the thermodynamic relation $\Delta H = E_a - RT$ ($RT \sim 2.48 \text{ kJ mol}^{-1}$) [28]. This difference reflects the definitions of the two parameters [27]. It confirms the consistency of the analysis over the temperature range 288–303 K. An increase in ΔH with increasing ethanol



SCHEME 2 | SWV curves for the $1 \cdot 10^{-3} \text{ mol dm}^{-3}$ Bi(III) ions electroreduction in (A) aqueous and (B–D) mixed aqueous-organic SEs containing ethanol at the concentrations indicated in the legend. Experimental conditions: frequency, 50 Hz; amplitude, 19.95 mV; potential step, 1.995 mV; starting potential, -0.3 V versus Ag/AgCl (3 M KCl); end potential, 0.2 V versus Ag/AgCl (3 M KCl) (SI, Table S5).

content was observed, suggesting enhanced stabilisation of the transition state in a less polar medium [29]. The lowest E_a ($10.72 \text{ kJ mol}^{-1}$) and ΔH (8.34 kJ mol^{-1}) values were recorded in the ethanol-free system (SE A), indicating the most favourable conditions for electron transfer in a highly polar, low-viscosity environment. The addition of 25 vol% ethanol (SE B) led to a notable increase in both E_a and ΔH , reaching maxima of 14.17 and $11.57 \text{ kJ mol}^{-1}$, respectively. Negative activation entropy values (ΔS), ranging from -281.81 to $-292.59 \text{ J} \cdot (\text{mol} \cdot \text{K})^{-1}$, suggest a high degree of ordering in the transition state and possible geometric constraints imposed by the presence of ethanol [30]. Despite differences in ΔH and ΔS , the Gibbs free energy of activation (ΔG)

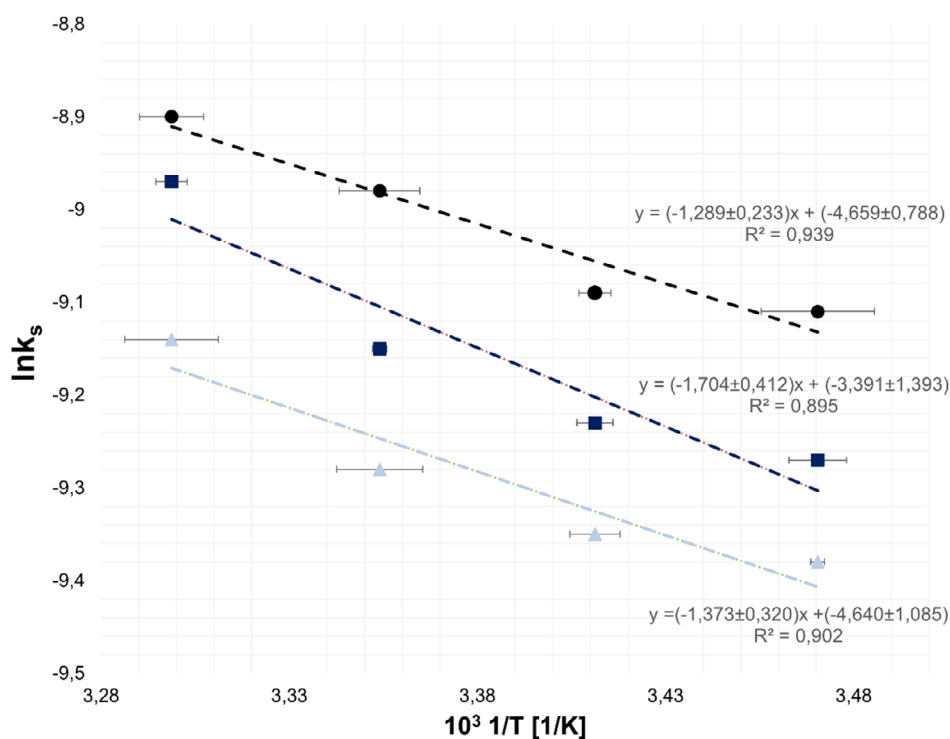
remained relatively constant ($\sim 95 \text{ kJ mol}^{-1}$), indicating enthalpy-entropy compensation [31]. Interestingly, in the 50% (v/v) ethanol system (SE C), ΔH decreased again to 8.71 kJ mol^{-1} , accompanied by the highest ΔS value, further supporting a mechanistic shift or altered interaction pattern that stabilises the transition state.

3.2 | Effect of Sodium 1-Octanesulfonate on the Bi(III) Ions Electroreduction: Interactions in the Context of the ‘Cap-Pair’ Effect

In the electroreduction of Bi(III) ions in solutions containing weakly complexing chlorate(VII) ions [32, 33], the ‘cap-pair’ rule

TABLE 2 | Formal potentials (E_f^0), reversible half-wave potentials ($E_r^{1/2}$) and cathodic transfer coefficients (α) for the $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(III) ions electroreduction in SEs.

T, K	E_f^0, V	$E_r^{1/2}, V$	α
0.5 mol dm $^{-3}$ NaClO $_4$ + 0.5 mol dm $^{-3}$ HClO $_4$ + $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(NO $_3$) $_3 \cdot 5H_2O$ (SE A)			
288	0.010	0.006	0.30
293	0.011	0.008	0.36
298	0.013	0.010	0.31
303	0.017	0.014	0.26
0.5 mol dm $^{-3}$ NaClO $_4$ + 0.5 mol dm $^{-3}$ HClO $_4$ + $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(NO $_3$) $_3 \cdot 5H_2O$ /25% EtOH (SE B)			
288	0.022	0.015	0.34
293	0.025	0.018	0.24
298	0.026	0.019	0.37
303	0.028	0.027	0.30
0.5 mol dm $^{-3}$ NaClO $_4$ + 0.5 mol dm $^{-3}$ HClO $_4$ + $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(NO $_3$) $_3 \cdot 5H_2O$ /50% EtOH (SE C)			
288	0.037	0.038	0.19
293	0.027	0.029	0.15
298	0.034	0.030	0.20
303	0.038	0.035	0.35



SCHEME 3 | Dependence of $\ln k_s$ on $1/T$ for the $1 \cdot 10^{-3}$ mol dm $^{-3}$ Bi(III) ions electroreduction in SEs (A–C) (0% $_{(v/v)}$) (●), 25% $_{(v/v)}$) (■), and 50% $_{(v/v)}$) (▼) EtOH). Trend lines were fitted using the Arrhenius equation; activation energies (E_a) were calculated from the slopes. Each k_s value was determined from three independent measurements; error bars represent the standard deviation of $\ln k_s$. The uncertainties of the fitted parameters were taken as the standard errors of the regression coefficients.

is often invoked to describe the influence of organic additives on the kinetics and mechanism of electrode processes [34, 35]. According to this concept, the presence of modifiers with specific structural features defined by the rule may result in three possible scenarios: (i) no effect on the kinetics, (ii) a catalytic effect-, e.g., through stabilisation of transient active complexes and

facilitation of charge transfer, or (iii) an inhibitory effect-, e.g., due to modifier adsorption and/or aggregate formation that physically hinders depolariser access to the electrode surface. The variability in the observed effects of the anionic surfactant 1OSASS across the studied systems indicates that the nature of the interactions depends on the SE composition. In the base solution

TABLE 3 | Thermodynamic activation parameters (E_a ; ΔH ; ΔS ; ΔG (at 298 K)) for the Bi(III) ions electroreduction in aqueous and aqueous-ethanol solutions of SE.

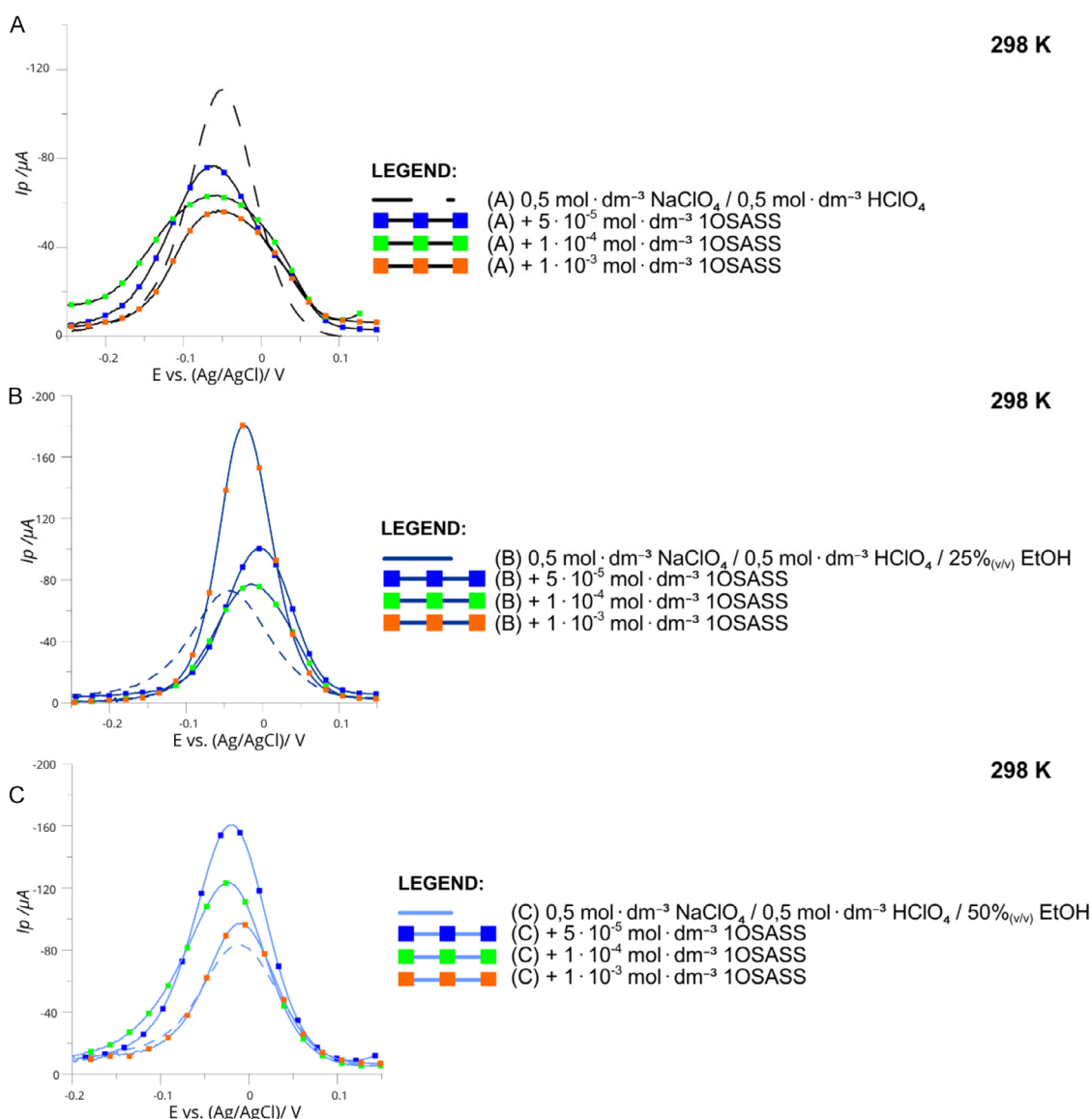
System	E_a , kJ mol^{-1}	ΔH , kJ mol^{-1}	ΔS , J (mol K)^{-1}	ΔG , kJ mol^{-1}
SE A	10.72	8.34	−291.61	95.24
SE B	14.17	11.57	−281.81	95.55
SE C	11.43	8.71	−292.59	95.90

containing $0.5 \text{ mol} \cdot \text{dm}^{-3} \text{ NaClO}_4 + 0.5 \text{ mol} \cdot \text{dm}^{-3} \text{ HClO}_4$ (SE A), the addition of 1OSASS results in a pronounced decrease in the reversibility of the Bi(III) electroreduction process (k_s), which is indicative of an inhibitory mechanism [36]. The systematic increase in the peak-to-peak separation (ΔE) between the anodic

and cathodic signals for Bi(III) ions electroreduction, observed via CV and presented in Table S2 (Supporting Information), further confirms the retardation of the electrode reaction. This is accompanied by a parallel decrease in the peak currents for Bi(III) ion electroreduction on the SWV curves (Scheme 4).

The observed changes are most likely due to the modified structure of surfactant-mediated formation of active complexes, as well as to the adsorption of 1OSASS molecules at the R-AgLAFE electrode/solution interface. This generally leads to a partial blockage of the active sites of the working electrode and a reorganisation of the EDL structure, which limits the efficiency of electron transfer and also reduces the intensity of the electrochemical response [14, 37].

In systems with ethanol (SE B and SE C), the influence of 1OSASS is changed. For lower surfactant concentrations, there is an increase and then a decrease in the peak currents in the



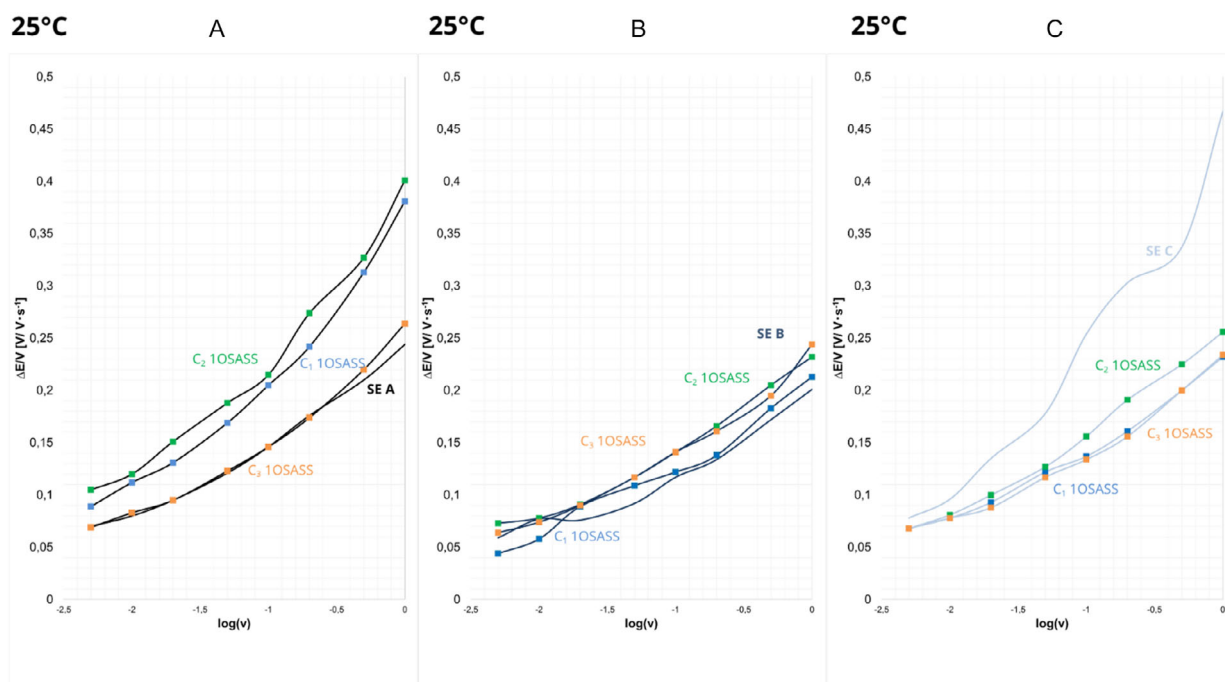
SCHEME 4 | SWV curves obtained at 298 K for the $1 \cdot 10^{-3} \text{ mol dm}^{-3}$ Bi(III) ions electroreduction in aqueous (A) and aqueous-organic (B,C) solutions of the SE containing ethanol at selected concentrations (indicated in the legend). Experimental conditions: electrolyte, chlorate(VII) solution (pH 0.25–0.30); frequency, 50 Hz; amplitude, 19.95 mV; potential step, 1.995 mV; starting potential, −0.2 V versus Ag/AgCl (3 M KCl); end potential, 0.2 V versus Ag/AgCl (3 M KCl) (SI, Table S5).

SWV curves relative to the peaks without the addition of 1OSASS (Scheme 4), which can be interpreted as a transitional catalytic effect. It most likely results from the stabilisation of the active complex structure in the presence of ethanol and surfactant. This is also confirmed by the ΔE values in SE B and SE C systems in the presence of 1OSASS compared to SE A (Supporting Information, Table S2–S4). For example, in system B, the ΔE values decrease from 0.074–0.323 V in the blank solution to 0.044–0.229 V at $5 \cdot 10^{-5}$ mol dm $^{-3}$ 1OSASS, while in system C they drop from 0.137–0.454 to 0.078–0.264 V under the same conditions (293 K). Significantly, it is only at higher concentrations of 1OSASS ($1 \cdot 10^{-3}$ mol dm $^{-3}$) that signs of secondary-inhibition appear, probably due to hemimicelle formation [38] (Scheme 5). The mechanism of this transient effect is consistent with known models: at low concentrations of surfactants (monomers), electron transfer is improved through co-adsorption and catalysis, while only above the CMC threshold does the formation of hemimicelles block the electrode surface and limit the efficiency of the electrochemical process [38, 39].

Analysis of the presented ΔE versus $\log(\nu)$ relationships (Scheme 5) further confirm that the effect of 1OSASS strongly depends on the type of solvent. In the case of SE A (Scheme 5A), each curve with surfactant addition lies higher than the reference curve (without addition), indicating an unambiguous inhibitory effect. In SE B (Scheme 5B), a partial overlap of the curves is observed. At lower concentrations (C_1), there is even a local reduction in ΔE relative to the base solution, confirming that the inhibitory effect is attenuated at lower electrode polarisation velocities. At SE C (Scheme 5C), all curves from 1OSASS are below the curve for SE C, clearly indicating the dominance of the catalytic effect in the range of scan rates and concentrations studied. This shows a clear shift in the mechanism of interactions from inhibitory (SE A) to transiently catalytic (SE B) and even

catalytic (SE C). This suggests that increasing the proportion of ethanol promotes EDL reorganisation and the formation of active complexes of depolariser-surfactant types, the structure of which facilitates charge transfer. However, this mechanism should be considered as dependent on simultaneous adsorption, electrostatic interactions, and solvent structure.

The determined thermodynamic parameters (Table 4) confirm previous considerations based on the ‘cap-pair’ effect mentioned earlier. The values obtained are consistent with observations of changes in ΔE and peak currents of Bi(III) ion electroreduction peaks in SWV curves. For the aqueous (chlorate(VII)) EP A system, the values of the enthalpy of activation ($\Delta H \sim 8.1$ – 9.2 kJ mol $^{-1}$) remain low and relatively stable. In contrast. The strongly negative entropy of activation ($\Delta S \sim -291$ to -296 J·(mol·K) $^{-1}$) indicates the ordered nature of the transition state. The increase in activation free energy (ΔG) with 1OSASS concentration correlates with the observed decrease in process reversibility and increase in ΔE , confirming the inhibition of the electrode process. In the EP B system (25%_(v/v) EtOH), the addition of 1OSASS causes more complex changes. For lower surfactant concentrations ($5 \cdot 10^{-5}$ mol dm $^{-3}$), a reduction in ΔH to 6.4 kJ mol $^{-1}$ and an even more negative ΔS (-298.7 J·(mol K) $^{-1}$) suggest transiently higher ordering of the system. Notably, the simultaneous decrease in ΔG may indicate transient stabilisation and a catalytic effect. At higher concentrations, ΔH increases (to more than 17 kJ·mol $^{-1}$), and ΔS becomes less damaging, indicating a change in mechanism and a possible transition to an activation energy-limited step, consistent with the observed secondary inhibition. The most pronounced changes in thermodynamic parameters were observed in the EP C (50% vol. EtOH) system. Both ΔH (an increase to 21.5 kJ mol $^{-1}$) and a gradual decrease in negative entropy (ΔS from -292 to -248 J·(mol K) $^{-1}$) indicate a marked change in the dynamics of the electrode process. The simultaneous reduction of ΔG and the



SCHEME 5 | Dependence of redox potential difference ΔE on logarithm of scan rate ($\log \nu$) for Bi(III) ions electroreduction ($C = 1 \cdot 10^{-3}$ mol dm $^{-3}$) at 298 K in the systems: (A) SE A (0%_(v/v) EtOH), (B) SE B (25%_(v/v) EtOH), as well as (C) SE C (50%_(v/v) EtOH), in non-additive solutions and in the presence of 1OSASS with concentrations of $C_1 = 5 \cdot 10^{-5}$, $C_2 = 1 \cdot 10^{-4}$, and $C_3 = 1 \cdot 10^{-3}$ mol dm $^{-3}$.

TABLE 4 | The values of the enthalpy of activation (ΔH), entropy of activation (ΔS), and free energy of activation (ΔG) for the electroreduction process of Bi(III) ions in SE A, SE B, and SE C systems and in the presence of additive 1OSASS at different concentrations. The calculations were made by interpreting the Eyring equation.

System	ΔH , kJ mol ⁻¹	ΔS , J(mol K) ⁻¹	ΔG , kJ mol ⁻¹
SE A	8.34	-291.61	95.24
SE A + 5·10 ⁻⁵ mol dm ⁻³ 1OSASS	8.11	-293.29	95.51
SE A + 1·10 ⁻⁴ mol dm ⁻³ 1OSASS	9.17	-291.16	95.94
SE A + 1·10 ⁻³ mol dm ⁻³ 1OSASS	8.34	-295.85	96.51
SE B	11.57	-281.81	95.55
SE B + 5·10 ⁻⁵ mol dm ⁻³ 1OSASS	6.44	-298.69	95.45
SE B + 1·10 ⁻⁴ mol dm ⁻³ 1OSASS	18.15	-257.41	94.86
SE B + 1·10 ⁻³ mol dm ⁻³ 1OSASS	17.63	-262.34	95.81
SE C	8.71	-292.59	95.90
SE C + 5·10 ⁻⁵ mol dm ⁻³ 1OSASS	10.51	-284.34	95.24
SE C + 1·10 ⁻⁴ mol dm ⁻³ 1OSASS	12.20	-265.96	91.45
SE C + 1·10 ⁻³ mol dm ⁻³ 1OSASS	21.45	-248.16	95.40

dominance of lower values of ΔE (Scheme 4) indicate a mechanism with catalytic features. It can be assumed that the presence of ethanol promotes EDL reorganisation and the formation of surfactant-depolariser type complexes with a structure that facilitates charge transfer. The mechanism is most likely based on the synergistic interactions among adsorption, solvation, and EDL reorganisation.

4 | Conclusion

The studies conducted enabled a kinetic and thermodynamic analysis of the Bi(III) ion electroreduction process in aqueous (chlorate(VII)) and aqueous-ethanol (chlorate(VII)-ethanol) systems, as well as in the presence of the surfactant 1OSASS, using an R-AgLAFe amalgam electrode. It was shown that both changes in the polarity of the reaction environment (ethanol content) and the presence of the surfactant significantly affect the electrical structure of the double layer, as well as the mechanism and kinetics of the redox process ('cap-pair' effect). Increasing ethanol content reduced the reversibility of the process. The influence of 1OSASS was strongly dependent on the system type—ranging from clear inhibition in aqueous solutions (EP A), to a mixed effect in a 25%(v/v) EtOH environment (EP B), and to pronounced catalysis in the presence of 50%(v/v) EtOH (EP C). Analysis of thermodynamic parameters (ΔH , ΔS , ΔG) revealed a correlation between the system's structure, the degree of transition-state ordering, and electron transfer efficiency. The results suggest that synergistic adsorption and solvation interactions are key factors shaping the Bi(III) electroreduction mechanism. The identified inhibitory/catalytic transition is a significant advantage of this study. Such effects may be crucial in systems where the structure of electrochemically deposited bismuth layers determines the functionality of heavy-metal-detection sensors. The analysis indicates that precise control of the electrode-reaction rate can be achieved by adjusting solvent polarity, thereby extending the traditional, often binary interpretation of interfacial effects as purely catalytic or inhibitory. The results demonstrate that polarity plays a critical role and provide a

foundation for further exploration of the 'cap-pair' effect. However, additional studies are necessary to fully understand the mechanisms underlying the observed behaviours.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. H. Yu., H. Guo, Y. Wang, Y. Wang, and L. Zhang, "Bismuth Nanomaterials as Contrast Agents for Radiography and Computed Tomography Imaging and Their Quality/Safety Considerations," *Wiley Interdisciplinary Reviews: Nanomedicine and Nanobiotechnology* 14 (2022): e1801.
2. C. Gomez, G. Hallot, S. Laurent, and M. Port, "Medical Applications of Metallic Bismuth Nanoparticles," *Pharmaceutics* 13 (2021): 1793.
3. T. Maeder, "Review of Bi₂O₃-Based Glasses for Electronics and Related Applications," *International Materials Reviews* 58 (2013): 3–40.
4. H. Liu, J. Zhao, T. Guo R. Sun, and X. Lu, "Nanolayered Bismuth Oxytelluride for Field-Effect Transistors and Photodetectors," *American Chemical Society* 8 (2025): 4100–4108.
5. P. Sivasubramanian, J. H. Chang, S. Nagendran, C. D. Dong, M. Shkir, and M. Kumar, "A Review on Bismuth-Based Nanocomposites for Energy and Environmental Applications," *Chemosphere* 307 (2022): 135652.
6. J. Wu, X. Kang, S. Xu, et al., "Influence of Bi³⁺ Doping on Electrochemical Properties of Ti/Sb-SnO₂/PbO₂ Electrode for Zinc Electrowinning," *Molecules (basel, Switzerland)* 29 (2024): 4062.
7. F. Yang, A. O. Elnabawy, and R. Schimmenti, "Bismuthene for Highly Efficient Carbon Dioxide Electroreduction Reaction," *Nature Communications* 11 (2020): 1088.
8. C. I. Fort, L. C. Cotet, A. Vulpoi, et al., "Bismuth Doped Carbon Xerogel Nanocomposite Incorporated in Chitosan Matrix for Ultrasensitive

- Voltammetric Detection of Pb(II) and Cd(II)," *Sensors and Actuators B: Chemical* 220 (2015): 712–719.
9. M. M. Rusu, C. I. Fort, L. C. Cotet, et al., "Insights into the Morphological and Structural Particularities of Highly Sensitive Porous Bismuth-Carbon Nanocomposites Based Electrochemical Sensors," *Sensors and Actuators B: Chemical* 268 (2018): 398–410.
 10. C. I. Fort, M. M. Rusu, L. C. Cotet, et al., "Carbon Xerogel Nanostructures with Integrated Bi and Fe Components for Hydrogen Peroxide and Heavy Metal Detection," *Molecule* 26 (2021): 117.
 11. K. Syku, G. Dalmata, B. Nowicka, and J. Saba, "Acceleration of Electrode Processes by Organic Compounds—cap-Pair Effect," *Journal of Electroanalytical Chemistry and Interfacial Electrochemistry* 90 (1978): 299–302.
 12. W. Kaliszczak and A. Nosal-Wiercińska, "Influence of Mixed 6-Thioguanine-Nonionic Surfactant Adsorption Layers on Kinetics and Mechanism of Bi(III) Ion Electroreduction," *Electrocatalysis* 10 (2019): 621–627.
 13. B. Timmer, M. Sluyters-Rehbach, and J. H. Sluyters, "Electrode Kinetics and Double Layer Structure," *Surface Science* 18 (1969): 44–61.
 14. W. R. Fawcett, "Fifty Years of Studies of Double Layer Effects in Electrode Kinetics—a Personal View," *Journal of Solid State Electrochemistry* 15 (2011): 1347–1358.
 15. Š. Komorsky-Lovrić, M. Lovrić, M. Branica, "Effect of Ionic Strength on Bi(III) Reduction from Chlorate(VII) Medium," *Journal of the Electrochemical Society* 140 (1993): 1850.
 16. A. Nosal-Wiercińska, M. Martyna, and A. Pawlak, "Application of the Nanostructured R-AgLAFE Electrode to Study the Electroreduction Process of Bi(III) Ions in the Presence of N-Acetylcysteine," *Applied Nanoscience* 13 (2023): 6811–6818.
 17. E. Sandnes, M. E. Williams, U. Bertocci, M. D. Vaudin, and G. R. Stafford, "Electrodeposition of Bismuth from Nitric Acid Electrolyte," *Electrochimica Acta* 52 (2007): 6221–6228.
 18. W. J. Li, "Electrodeposition of Bismuth Telluride Films from a Nonaqueous Solvent," *Electrochimica Acta* 54 (2009): 7167–7172.
 19. A. Nosal-Wiercińska, M. Martyna, V. Mirčeski, and S. Skrzypek, "Electroreduction of Bi(III) Ions at a Cyclically Renewable Liquid Silver Amalgam Film Electrode in the Presence of Methionine," *Molecules (basel, Switzerland)* 26 (2021): 3972.
 20. A. Nosal-Wiercińska, M. Martyna, M. Grochowski, and B. Baś, "First Electrochemical Studies on CAP—PAIR Effect for Bi(III) Ion Electroreduction in the Presence of 2-Thiocytosine on Novel Cyclically Renewable Liquid Silver Amalgam Film Electrode (R-AgLAFE)," *Journal of the Electrochemical Society* 168 (2021): 066504.
 21. Z. Galus, "Influence of Solvents on the Rate of Electrode Reactions," *Pure and Applied Chemistry* 63 (1991): 1705–1714.
 22. J. Lipkowski and Z. Galus, "Electrode Kinetics in Mixed Water + Aliphatic Alcohol Solvents," *Journal of Electroanalytical Chemistry and Interfacial Electrochemistry* 58 (1975): 51–69.
 23. United States Pharmacopeial Convention (USP), "Monograph c801". In USP 32–NF 27. Can Be Found under, 2009, Available at: https://www.drugfuture.com/pharmacopoeia/usp32/pub/data/v32270/usp32nf27s0_c801.html (accessed: 12 VII 2025).
 24. Z. Galus, *Elektroanalityczne Metody Wyznaczania Stałych Fizykochemicznych* (PWN, 1979).
 25. R. Chang, *Physical Chemistry for the Biosciences* (University Science Books, 2005), 338–342.
 26. , "Transition State Theory: Eyring Equation", LibreTexts Chemistry, Can Be Found under (accessed: 12 VII 2025), Available at: https://chem.libretexts.org/.../Eyring_equation.
 27. R. R. Krug, W. G. Hunter, and R. A. Grieger, "Enthalpy-Entropy Compensation. 2. Separation of the Chemical from the Statistical Effect," *Journal of Physical Chemistry* 80 (1976): 2341–2351.
 28. J. Guerra, B. Bajwa, P. Kumar, S. Vazquez, V. V. Krishnan, and S. Maitra, "Validation of Enthalpy-Entropy Compensation Mechanism in Partial Amide Bond Rotation," *ACS Omega* 5 (2020): 9348–9355.
 29. A. K. Singh, "Solvent Effect on the Enthalpy and Entropy of Activation for the Hydrolysis of Ethyl Cinnamate in Mixed Solvent System," *Journal of Physical Chemistry Biophysics* 7 (2017): 238.
 30. J. D. Chodera and D. L. Mobley, "Entropy-enthalpy Compensation: Role and Ramifications in Biomolecular Ligand Recognition and Design," *Annual Review of Biophysics* 42 (2013): 121–142.
 31. A. Pan, T. Biswas, A. K. Rakshit, and S. P. Moulik, "Enthalpy-Entropy Compensation (EEC) Effect: A Revisit," *Journal of Physical Chemistry B* 119 (2015): 15876–15884.
 32. M. R. Rosenthal, "The Myth of the Non-Coordinating Anion," *Journal of Chemical Education* 50 (1973): 331.
 33. S. Saha, S. Pramanik, S. Pathak, et al., "Noncoordinating Anions as Key Modulators of Supramolecular Structures, Optical and Electrical Properties in Nickel(II) Complexes," *ACS Omega* 9 (2024): 44494–44506.
 34. J. Nieszporek, T. Pańczyk, and K. Nieszporek, "The Comparison of Catalytic Activity of Carbimazole and Methimazole on Electroreduction of Zinc(II) in Chlorates(VII): Experimental and Molecular Modelling Study," *Molecules* 29 (2024): 3455.
 35. J. Nieszporek, D. Gugała-Fekner, and K. Nieszporek, "The Effect of Supporting Electrolyte Concentration on Zinc Electrodeposition Kinetics from Methimazole Solutions," *Electroanalysis* 31 (2019): 1141.
 36. A. Pawlak and A. Nosal-Wiercińska, "Effect of Ionic Surfactants on Kinetics and Mechanism of the Bi(III) Ion Electroreduction in the Mixed Aqueous–Organic Solutions of Supporting Electrolytes," *Molecules* 29 (2024): 4986.
 37. A. H. B. Dourado, "Electric Double Layer: The Good, the Bad, and the Beauty," *Electrochem* 3 (2022): 789–808.
 38. R. Vittal, H. Gomathi, and K.-J. Kim, "Beneficial Role of Surfactants in Electrochemistry and the Modification of Electrodes," *Advanced Colloid and Interface Science* 119 (2006): 55–68.
 39. K. Wang, Y. Wang, and M. Pera-Titus, "Liquid-Liquid and Gas-Liquid Dispersions in Electrochemistry: Concepts, Applications and Perspectives," *Chemical Society Reviews* 53 (2024): 11701–11724.